1 Introduction

The purpose of laboratory is observing application of Kirchhoff's Current Law and Kirchhoff's Voltage Law *(we will state them as KCL and KVL after that)* and simulating these calculations on Circuit Lab simulation environment to see if our calculations are correct.

1.1 Preliminary Work

Before using Circuit Lab, we apply KCL and KVL to task 0.

- First of all, we calculate KCL on each node on all over the tasks.
- Then we calculate KVL on each loop on all over the tasks.
- Finally we take notes of our currents and voltages from our calculations to compare with simulation results later.

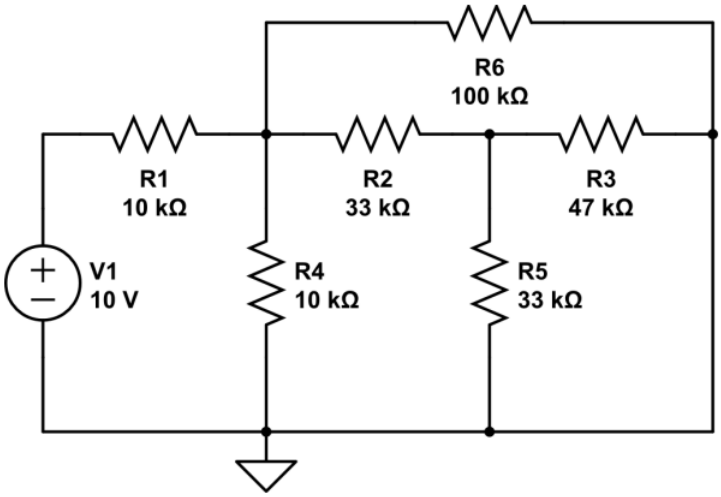


Figure 1: Task 1.

Calculation of task 0 for KCL is on the next page.

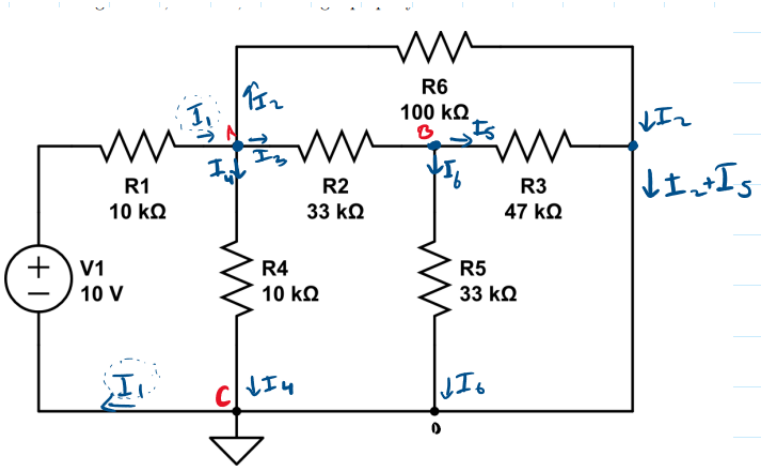


Figure 2: Current representation on circuit Task 0.

Table 1: Table that shows KCL calculations. Currents = Equations.

I	*Equations*
I_1	$I_2 + I_3 + I_4$
I_3	$I_5 + I_6$
I_1	$I_2 + I_4 + I_5 + I_6$

Calculation of task 0 for KVL is on the next page.

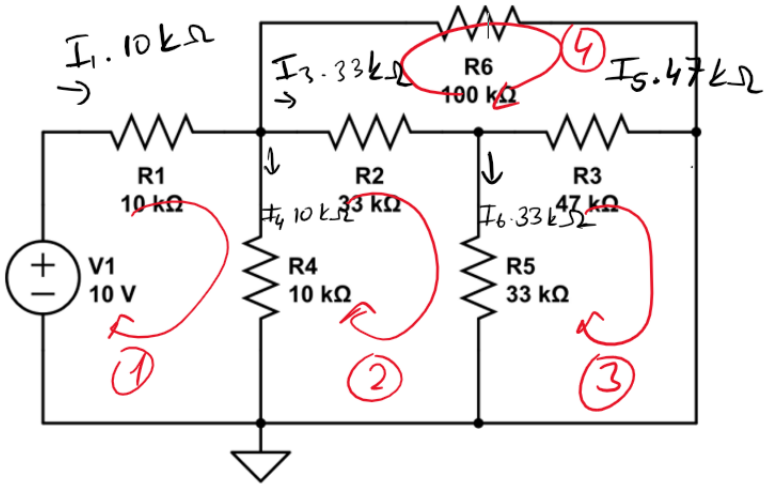


Figure 3: Voltage loops representation on circuit Task 0.

Table 2: Table that shows KVL calculations. Voltages = Equations.

Voltages	*Equations*
$10V$	$I_1 10 + I_4 10$
I_4	$I_3 33 + I_6 33$
$I_2 100$	$I_3 33 + I_5 47$
$I_6 33$	$I_5 47$

With calculation of KCL and KVL, we obtain a matrix which is presented on next page:

Solution of that matrix is on next page,

Table 3: Matrix representation of solution with KCL and KVL.

-10000	0	0	-10000	0	0	**10**
0	0	-33000	10000	0	-33000	**0**
0	0	0	0	-47000	33000	**0**
0	-100000	33000	0	47000	0	**0**
-1	1	1	1	0	0	**0**
0	0	-1	0	1	1	**0**

Table 4: **If we solve that matrix we find out that voltages for these resistors.**

R_1	$5,6V$
R_2	$2,7V$
R_3	$1,6V$
R_4	$4,4V$
R_5	$1,6V$
R_6	$4V$

Simulation results of task0 is on the next page.

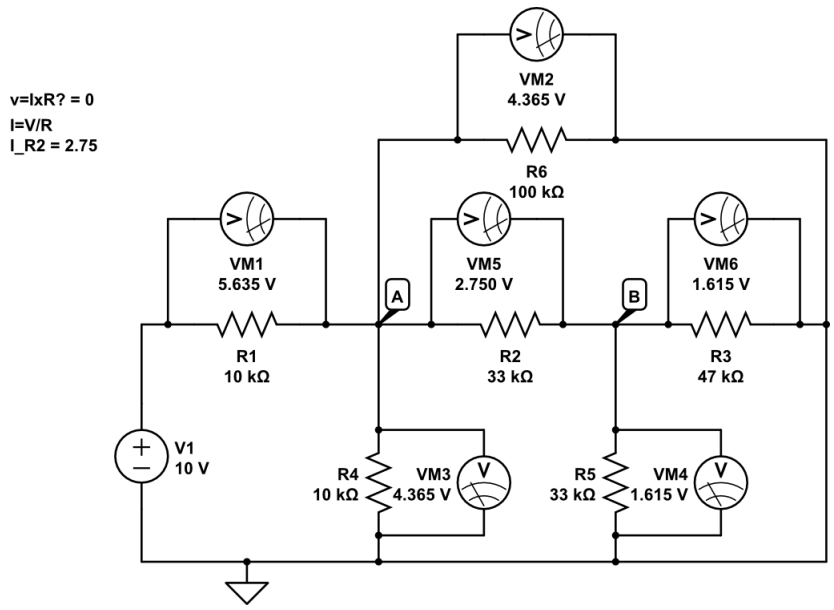


Figure 4: The simulation of Circuit Lab with task0.

1.2 Conclusion of Preliminary work

As we can see from KVL and KCL conclusions, and by the comparison with Circuit Lab results. We can understand our KCL and KVL calculations are correct. Here is the link of simulation: (https://www.circuitlab.com/circuit/jnbgx4zg4yz9/lab1$_c ircuit1$/)

1.3 Lab Tasks

$OHMSLAW => V = I * R$ **That equation represents Voltage equals to Current times Equivalent Resistance of system.** In Lab01 we are building 3 different circuits with different resistors but same power supply which is fixed on 1kHz. In that experiment we are looking to find relationship between resistor resistances and amperes and volts. To do that we are looking ampere and voltage graphics of circuit on time domain.

1.3.1 Task 1.

1. First, constructed the circuit task1 *(Which is shown on next page)* on Circuit Lab, we are measuring the voltage with plugging voltmeters.

2. circuit task1 has 1 power source and 2 resistors with 47 ohm, also has two nodes named as **A** and **B**.

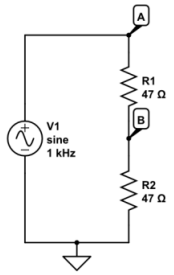


Figure 5: Task 1 on Circuit Lab simulation.

After running Time Domain Simulation on Circuit Lab we obtain that information on Table 5 above which presents the graph of task 1 values on next page: In task 1 we have double voltage and ampere difference between two nodes because we have two same amount of resistance on circuit. From ohm's law we have total 94 because two 47 ohm resistors are connected in series. 94 ohm resistance and 1kHz voltage source comes from power supply, which makes our current 1000/94=Current= 10.63 as exactly same with simulation. Here is the simulation link : (https://www.circuitlab.com/circuit/wqk9u6n2saq8/lab1$_c ircuit$2/)

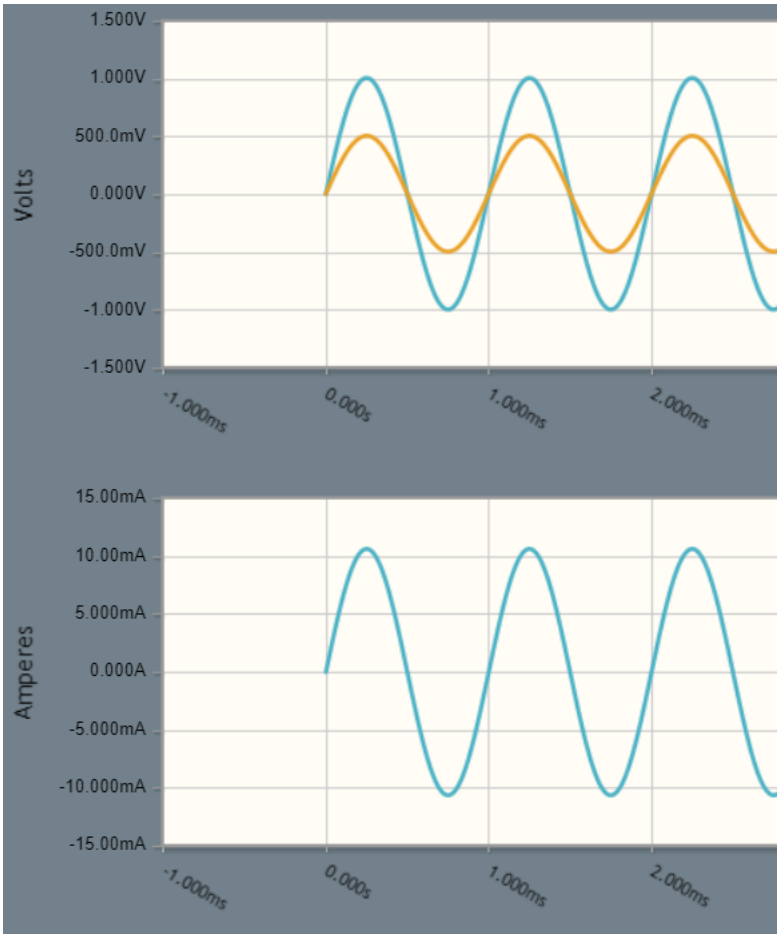


Figure 6: Graph of Task 1.

1.3.2 Task 2.

1. First, constructed the circuit task2.b *(Which is shown below)*on Circuit Lab, we are measuring the voltage with plugging voltmeters.

2. circuit task2.a has 1 power source and 2 resistors with 1k ohm, also has two nodes named as **C** and **D**. The circuit figure will be on next page.

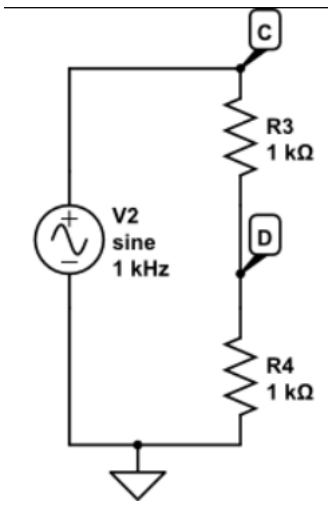


Figure 7: task 2 on Circuit Lab simulation.

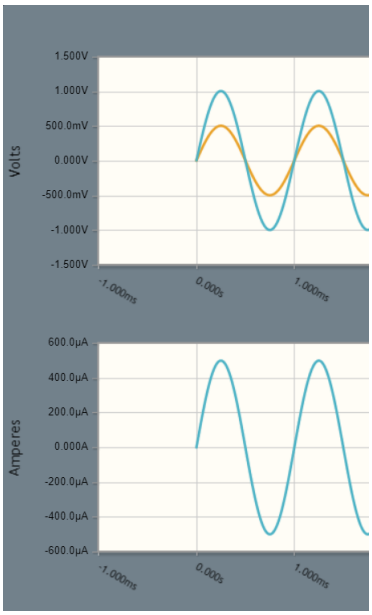


Figure 8: Graph of task 2.

Numeric explanation will be on next page.

Table 6: Table for numeric explanation of the graphic on Figure 8 which is on previous page.

µs	V	µAmp
250µs	1000V	500
500µs	0V	0
750µs	-1000V	-500

After running Time Domain Simulation on Circuit Lab we obtain that information on Table 6 above which presents the graph of task 2 values: In task 2 we have double voltage and ampere difference between two nodes because we have two same amount of resistance on circuit. From ohm's law we have total 2k ohm resistance because two 1k ohm resistors are connected in series and 1kHz voltage source comes from power supply, which makes our current 1000/2000=Current= 0.500 as exactly same with simulation. Here is the simulation link: (https://www.circuitlab.com/circuit/wqk9u6n2saq8/lab1$_c ircuit$2/)

1.3.3 Task 3.

1. First, constructed the circuit task2.b *(Which is shown below)*on Circuit Lab, we are measuring the voltage with plugging voltmeters.

2. circuit task2.a has 1 power source and 2 resistors R5 is 500k ohm and R6 is 1.5M ohm, also has two nodes named as **E** and **F**. The circuit figure will be on next page.

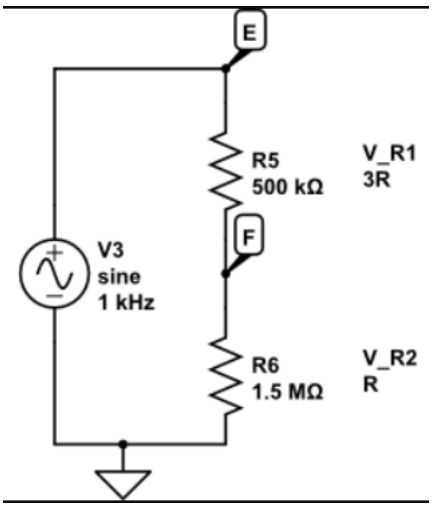


Figure 9: task 3 on Circuit Lab simulation.

Graph and numeric explanation will be on next page. After running Time Domain Simulation

Table 7: Table for numeric explanation of the graphic on Figure 10 which is below

µs	V	nAmp
250µs	1000V	500
500µs	0V	0
750µs	-1000V	500

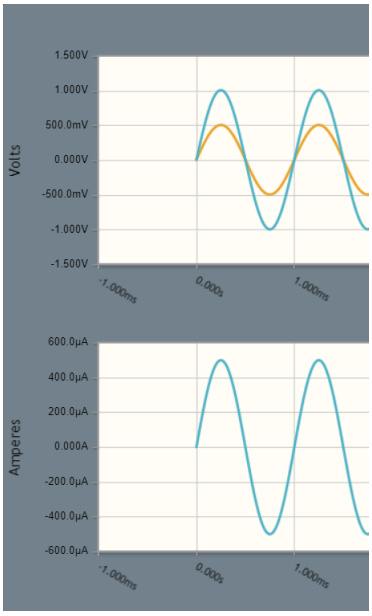


Figure 10: Graph of task 3.

on Circuit Lab we obtain that information on Table 7 above which presents the graph of task 3 values: In task 3 we have different voltage and ampere difference from task 2 and task 1 on between two nodes because we have two different amount of resistance on circuit. Now the ratio of resistors resistance is 1/3. Voltage difference is same as the ratio. From ohm's law we have 1.5M and 0.5M ohm resistance which are they connected in series so that gives us equivalent resistance which is 2M and 1kHz voltage source comes from power supply, which makes our current 1000/2m=Current= n500 as exactly same with simulation. Here is the simulation link: (https://www.circuitlab.com/circuit/wqk9u6n2saq8/lab1$_circuit$2/)

2 Conclusion

In conclusion, we have done preliminary work to understand the concept at first with more complicated circuit than task1, task2 and task3. In our tasks all of the resistors are connected in series, so that effects the voltage and ampere passes through resistors, we have have built the circuits on Circuit Lab Simulation program and simulated to check their values.